
Neuro-Inspired Computing Based on a FPAA Platform

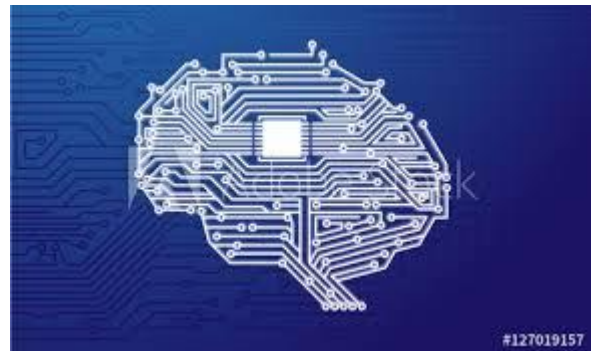
- Spring 2026- undergraduate research/special problem (1, 2 or 3 SCH)

Supervisor: Dr. Alexandre Locquet

An interested student should send an email to Dr. Locquet and Dr. Hasler expressing interest (Emails: jennifer.hasler@ece.gatech.edu and alexandre@gatech.edu) or stop by Dr. Locquet's (#206) or Hasler's office.

Context:

Similar to FPGAs, field-programmable analog arrays (FPAAs) are arrays of electronic gates but the difference lies in that they incorporate a large number of analog components. The use of analog components can allow for easier interfacing with external physical signals and lead to significantly improved energy efficiency. The Georgia Tech-CNRS lab makes use of a FPAA platform from Dr. Hasler' group to benchmark implementations of neuro-inspired computing. Neuro-inspired computing is a new form of computing that tries to leverage the computational processes taking place in the brain and not to implement traditional concepts from Turing and Von Neumann's machines. Neuro-inspired architectures involves arrays of interconnected nodes, also called artificial neurons, which are nonlinear dynamical systems. These architectures can be efficient at solving complex tasks such as nonlinear times-series prediction or optimization problems (*e.g.* traveling salesman, max-cut)



Objectives:

The objective is to build the basic constitutive elements of a neuro-inspired computer on the FPAA platform.

The students will contribute to the following tasks

- (1) Implement the basic building blocks of a neuron, in particular: nonlinearities, filters, delay line.
- (2) Determine and implement nonlinear dynamical systems that could serve as neurons.
- (3) Determine a framework for interconnecting various neurons and realize a network.

References / Resources:

- [1] <https://irl2958.georgiatech-europe.eu/terahertznde-nonlineardynamicsgroup/> for information on nonlinear dynamical systems
- [2] <http://hasler.ece.gatech.edu/> for information on FPAAs

Procedure to register for the special problem/undergraduate research:



Course number:

For 3SCH: ECE2699 if freshman/sophomore, ECE4699 if junior/senior

For 2SCH: ECE 1902/2902/3902/4902 depending on freshman/sophomore/junior/senior status

For 1SCH: ECE 1901/2901/3901/4901 depending on freshman/sophomore/junior/senior status

Students need to submit a request to register for undergraduate research or undergraduate special problem: <https://www.ece.gatech.edu/ece-undergraduate-docusign-forms>